

MANGALORE INSTITUTE OF TECHNOLOGY & ENGINEERING

(A Unit of Rajalaxmi Education Trust ®, Mangalore)

Autonomous Institute Affiliated to V.T.U, Belagavi, Approved by AICTE, New Delhi

Accredited by NAAC with A+ Grade & ISO 9001:2015 Certified Institution



Real-Time Sign Language Alphabet Recognition Using MobileNetV2 Transfer Learning

**A thesis report submitted in partial fulfillment of the requirements for
the award of degree of**

MASTER OF COMPUTER APPLICATIONS

Submitted by

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DEPARTMENT OF MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that the thesis work entitled “**REAL-TIME SIGN LANGUAGE ALPHABET RECOGNITION USING MOBILENETV2 TRANSFER LEARNING**” is a bonafide work carried out by **AKSHAY G (4MT24MC003)**, in partial fulfillment for the award of degree of **Master of Computer Applications of Mangalore Institute of Technology & Engineering (Autonomous)** during the year 2025-26. It is certified that all corrections and suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The thesis has been approved as it satisfies the academic requirements in respect of Major Project Phase-2 work prescribed for the Master of Computer Applications degree.

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ABSTRACT

This thesis presents the design, implementation, and evaluation of a Real-Time Sign Language Alphabet Recognition Using MobileNetV2 Transfer Learning, an automated translation tool engineered to interpret static hand postures representing the twenty-nine core characters of the English manual alphabet and numerals. Bridging the communication gap between the deaf community and the hearing majority often requires expensive hardware or human interpreters. To address this, the proposed system leverages the computationally efficient MobileNetV2 convolutional neural network via transfer learning, operating entirely on standard consumer webcams without the need for specialized sensors. The classification model is integrated into a lightweight, browser-based Flask web application, facilitating real-time inference and seamless user interaction. By isolating a bounding box from the video feed and continuously classifying the captured frames, the system allows users to intuitively form complete words. Empirical evaluation demonstrates high predictive accuracy and low latency on commodity hardware, validating the system as an accessible, cost-effective assistive technology for real-time sign language translation.

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